

Observations for Structural Simulation (FEA) of Cantilever Beam

Case 1: Load = 1000 N, Fixed Support

1. The maximum deformation occurs at the free end of the beam.
2. The deformation gradually decreases from the free end towards the fixed support.
3. Maximum von Mises stress is observed near the fixed support region.
4. Stress concentration occurs at the beam-root where the beam is constrained.
5. The factor of safety is greater than 1, indicating that the beam is safe under the applied load.
6. The stress value is below the yield strength of structural steel, so no permanent deformation is expected.

Case 2: Load = 2000 N, Fixed Support

1. The total deformation approximately doubles compared to the 1000 N load case.
2. The maximum stress increases significantly with the increase in load.
3. The highest stress remains concentrated near the fixed support.
4. The factor of safety decreases due to increased loading.
5. If the stress exceeds the yield strength of the material, plastic deformation or failure may occur.
6. The beam becomes more susceptible to structural failure under higher loads.

Effect of Boundary Conditions

1. Fixed support provides maximum restraint against movement and rotation.
2. The support condition strongly influences stress distribution and deformation.
3. A fixed-fixed beam experiences lower deformation than a cantilever beam under the same load.
4. Proper boundary condition selection is essential for obtaining realistic simulation results.

Mesh Observation

1. The mesh divides the beam into small finite elements for numerical analysis.

2. A finer mesh improves result accuracy but increases computational time.
3. Stress values near support regions become more accurate with mesh refinement.
4. The generated mesh was sufficiently fine to capture stress concentration effects.

Stress Distribution Observation

1. Stress is not uniformly distributed throughout the beam.
2. Maximum stress occurs at the fixed end due to the highest bending moment.
3. Stress decreases gradually along the beam length toward the free end.
4. The stress contour clearly indicates critical regions that require design attention.

Deformation Observation

1. Deformation follows the bending shape expected for a cantilever beam.
2. The free end undergoes the largest displacement.
3. Deformation increases proportionally with applied load.
4. Structural stiffness directly affects the magnitude of deformation.

Factor of Safety Observation

1. The factor of safety decreases as applied load increases.
2. Regions near the fixed support show the minimum factor of safety.
3. A factor of safety greater than 1 indicates safe operation.
4. A factor of safety less than 1 indicates possible yielding or structural failure.

Overall Conclusion

The FEA simulation successfully predicted the behavior of the cantilever beam under different loading conditions. The results showed that stress concentration occurs near the fixed support, while maximum deformation occurs at the free end. Increasing the applied load resulted in higher stresses and deformations and reduced the factor of safety. The simulation

demonstrates the importance of load magnitude, material properties, and boundary conditions in structural design and analysis.